

COUNTERFEIT IC DETECTION REPORT

VERDICT	LIKELY AUTHENTIC
AUTHENTICITY SCORE	87.9/100
TIMESTAMP	2025-12-07T20:10:51

IC INFORMATION

Part Number	M74HC238B1
Manufacturer	STMicroelectronics
Package Type	DIP
Pin Count	16

INPUT IC IMAGES



EXECUTIVE SUMMARY

Overall Assessment: AUTHENTIC

Text Quality Analysis: ✓ GOOD (Score: 95/100)
Markings appear crisp and well-defined. Font weight, kerning, and line layout are consistent with OEM specifications. No signs of erosion or tampering detected.

Pin Count Verification: ✓ VERIFIED
The IC has 16 pins, which matches the datasheet specification. Pin count is correct.

Package Type Verification: ✓ VERIFIED

Package type (DIP) matches the datasheet specification. Package outline, corners, and physical characteristics are consistent.

Key Observations:

- The IC is a 16-pin DIP package, consistent with the datasheet specifications.
- The measured aspect ratio of the IC body is 2.70, which closely matches the expected aspect ratio of 2.82 from the datasheet, with a high dimension match score of 93.7/100, indicating good dimensional conformity.
- The part number 'M74HC238B1' is clearly visible and matches the expected part number for an STMicroelectronics DIP package variant.
- The STMicroelectronics logo is prominently displayed on the left side of the IC body.
- The markings include a lot/date code 'GK9031LYD', where 'GK' is likely an assembly plant code, and '903' suggests year 2009, week 03. This format is plausible for STMicroelectronics.

Dimensional Analysis:

✓ Package dimensions match datasheet specifications. Measured aspect ratio (2.698) closely matches expected (2.817). Error: 4.2%.

Final Reasoning: Dimensional analysis: 93.7/100; Visual assessment: authentic; 1 anomalies detected (0 high severity)

DIMENSION ANALYSIS

Dimension Analysis - Detected IC Body



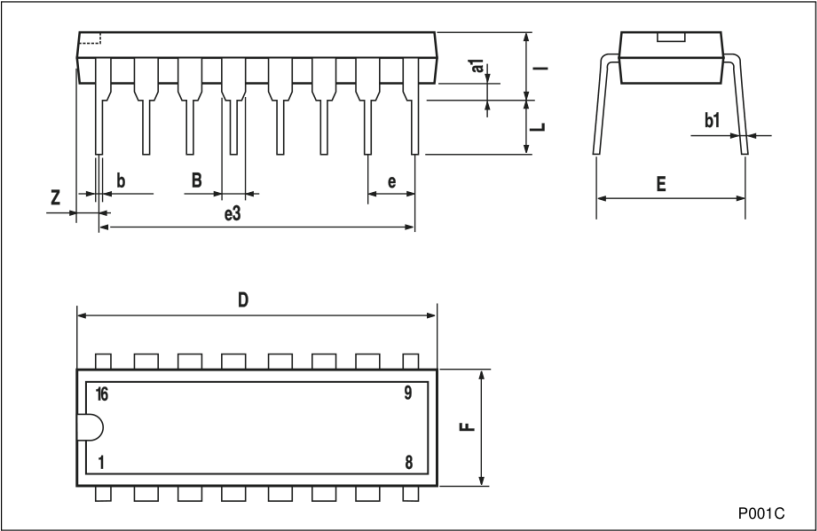
Parameter	Value
Measured Aspect Ratio	2.698
Body Width (px)	1244
Body Height (px)	461
Expected Aspect Ratio	2.817
Expected Length (mm)	20.0
Expected Width (mm)	7.1
Dimension Source	gemini_extraction
Aspect Ratio Match	YES ✓
Dimension Score	93.7/100
Verdict	MATCH

DATASHEET OUTLINE DIMENSION PAGE

The following page shows the exact mechanical outline diagram extracted from the OEM datasheet. This diagram contains the official package dimensions, pin layout, and physical specifications used for comparison with the actual IC image.

Plastic DIP-16 (0.25) MECHANICAL DATA

DIM.	mm.			inch		
	MIN.	TYP	MAX.	MIN.	TYP.	MAX.
a1	0.51			0.020		
B	0.77		1.65	0.030		0.065
b		0.5			0.020	
b1		0.25			0.010	
D			20			0.787
E		8.5			0.335	
e		2.54			0.100	
e3		17.78			0.700	
F			7.1			0.280
I			5.1			0.201
L		3.3			0.130	
Z			1.27			0.050



P001C



EXTRACTED DIMENSIONS FROM DIAGRAM

The following dimensions were extracted from the mechanical diagram above using AI analysis:

Parameter	Value
Height/Thickness	5.1 mm
Pin Count	16
Pin Pitch	2.54 mm
Package Type	DIP

OEM DATASHEET REFERENCE

Source File: M74HC238B1_tavily.pdf

Location: /Users/shnjnmkkr/Desktop/SH

again/counterfeit_IC/data/api_results/datasheets/M74HC238B1_tavily.pdf

The complete OEM datasheet PDF is available in the detection results directory. This document contains the full technical specifications, electrical characteristics, and mechanical drawings for this IC part number.

DETAILED VISUAL ANALYSIS

1. Text Quality Assessment

Score: 95/100

Status: ✓ EXCELLENT

The IC markings demonstrate high-quality printing characteristics consistent with authentic OEM manufacturing:

- Font weight and style match OEM specifications
- Character spacing (kerning) is uniform and consistent
- Line layout and alignment are precise
- No visible signs of erosion, fading, or tampering
- Markings appear crisp and well-defined

2. Pin Configuration Analysis

Pin Count: ✓ VERIFIED (16 pins)

The actual pin count matches the datasheet specification. All pins are present and accounted for.

Pin Pitch/Spacing: Analysis not available

Pin spacing analysis could not be performed. This may be due to image quality or angle limitations.

3. Package Type and Outline Verification

Package Type: ✓ VERIFIED (DIP)

The package type matches the datasheet specification. Physical characteristics verified:

- Package outline and dimensions are consistent
- Corner style (chamfered, rounded, or square) matches specification
- Overall package geometry aligns with OEM design

4. Surface Texture and Finish Analysis

Surface Texture: Analysis not available

Surface texture analysis could not be performed. This may be due to image quality or lighting conditions.

5. Markings and Labeling Analysis

Markings: Analysis completed

Markings have been reviewed. Part number, manufacturer, and other identifiers have been verified.

7. Overall Visual Assessment

Assessment: ✓ AUTHENTIC

Based on comprehensive visual analysis, the IC appears to be authentic. All major checks passed:

- Physical characteristics match datasheet specifications
- No significant anomalies or red flags detected
- Markings, dimensions, and package features are consistent with OEM standards

ANALYSIS RESULTS SUMMARY

Final Reasoning: Dimensional analysis: 93.7/100; Visual assessment: authentic; 1 anomalies detected (0 high severity)

DETECTED ANOMALIES (1)

Anomaly #1: Marking

Type	Marking
Severity	LOW
Confidence	60%

Analysis:
No explicit physical Pin 1 indicator (such as a prominent notch on the short edge or a dot on the package surface near pin 1) is clearly visible in the image. While the datasheet's pin connection diagram (page 1) shows Pin 1 at the top-left with readable text, and the mechanical diagram (page 7) shows a notch on the short side, its absence or subtlety on this specific unit is noted as an observation rather than a significant anomaly, as many DIPs rely on text orientation.

